Section 2 - Practice Problems

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October 21, 2025

Chapter 2.1: The Algebraic and Order Properties of $\mathbb{R}$

Exercise 2.1.1 Solve the following equations, justifying each step by referring to an appropriate property or theorem.

(a) $2x + 5 = 8$.

Solution:

$$\begin{aligned}
2x + 5 = 8 &\implies \\
2x + 5 - 5 = 8 - 5 &\implies && \langle \text{By the additive property of equality} \rangle \\
2x + 0 = 3 &\implies && \langle \text{By the additive inverse property} \rangle \\
2x = &\implies && \langle \text{By the additive identity property} \rangle \\
\frac{2x}{2} = \frac{3}{2} &\implies && \langle \text{By the multiplicative property of equality} \rangle \\
\frac{2}{2} \cdot x = \frac{3}{2} &\implies && \langle \text{By the associativity of multiplication} \rangle \\
1 \cdot x = \frac{3}{2} &\implies && \langle \text{By the multiplicative inverse property} \rangle \\
x = \frac{3}{2} & && \langle \text{By the multiplicative identity property}
\end{aligned}$$

(b) $x^2 = 2x$

Solution:

$$
\begin{aligned}
x^2 = 2x &\implies \\
x^2 - 2x = 2x - 2x &\implies && \langle \text{By the additive property of equality} \rangle \\
x^2 - 2x = 0 &\implies && \langle \text{By the additive inverse property} \rangle \\
x \cdot (x - 2) = 0 &\implies && \langle \text{By the distributive property} \rangle \\
x = 0 \text{ or } x = 2 & && \langle \text{By Theorem 2.1.3.b} \rangle
\end{aligned}
$$

(c) $x^2 - 1 = 3$

Solution:

$$
\begin{aligned}
x^2 - 1 = 3 &\implies \\
x^2 - 1 - 3 = 3 - 3 &\implies && \langle \text{By the additive property of equality} \rangle \\
x^2 - 4 = 0 &\implies && \langle \text{By the additive inverse property} \rangle \\
(x + 2) \cdot (x - 2) = 0 &\implies && \langle \text{By the distributive property} \rangle \\
x = 2 \ \vee \ x = -2 & && \langle \text{By Theorem 2.1.3.a} \rangle
\end{aligned}
$$

(d) $(x-1)(x-2)=0$

Solution:

$$(x-1)(x-2)=0 \implies$$
$$x=1 \lor x=2 \quad \langle \text{By Theorem 2.1.3.a} \rangle$$

Exercise 2.1.9 Let $K \coloneqq \{s+t\sqrt{2} : s,t \in \mathbb{Q}\}$. Show that K satisfies the following:

(a) If $x_1, x_2 \in K$, then $x_1+x_2 \in K$ and $x_1x_2 \in K$.

Solution:

Let $x_1 = s_1 + t_1\sqrt{2}, x_2 = s_2 + t_2\sqrt{2} \in K$. We separate the proof of each claim into two parts into two parts.

(I) Prove $x_1 + x_2 \in K$

Proof. We can see that:

$$\begin{aligned} x_1 + x_2 &= s_1 + t_1\sqrt{2} + s_2 + t_2\sqrt{2} \\ &= (s_1+s_2) + (t_1+t_2)\sqrt{2} \end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under addition, we know that $(s_1+s_2) \in \mathbb{Q}$ and $(t_1+t_2) \in \mathbb{Q}$. Therefore, $((s_1+s_2)+(t_1+t_2)\sqrt{2}) \in K$ and thus $(x_1+x_2) \in K$. □

(II) Prove $x_1x_2 \in K$

Proof. We can see that:

$$\begin{aligned} x_1x_2 &= (s_1 + t_1\sqrt{2})(s_2 + t_2\sqrt{2}) \\ &= s_1s_2 + s_1t_2\sqrt{2} + s_2t_1\sqrt{2} + t_1t_2\sqrt{2}\sqrt{2} \\ &= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + t_1t_2(\sqrt{2})^2 \\ &= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + 2t_1t_2 \\ &= (s_1s_2 + 2t_1t_2) + (s_1t_2 + s_2t_1)\sqrt{2} \end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under both addition and multiplication, we know that $s_1s_2+2t_1t_2 \in \mathbb{Q}$ and $s_1t_2+s_2t_1 \in \mathbb{Q}$. Therefore, $(s_1s_2+st_1t_2)+(s_1t_2+s_2t_1)\sqrt{2} \in K$ and thus $x_1x_2 \in K$. □

(b) If $x \neq 0$ and $x \in K$, then $\frac{1}{x} \in K$.

Solution:

Proof. Let $x = s + t\sqrt{2} \in K$, where s and t cannot equal 0 simultaneously. Therefore, we have:

$$\begin{aligned}
\frac{1}{x} &= \frac{1}{s+t\sqrt{2}} \\
&= \frac{1}{s+t\sqrt{2}} \cdot \frac{s-t\sqrt{2}}{s-t\sqrt{2}} \\
&= \frac{s-t\sqrt{2}}{(s+t\sqrt{2})(s-t\sqrt{2})} \\
&= \frac{s-t\sqrt{2}}{s^2+(t\sqrt{2})^2} \\
&= \frac{s-t\sqrt{2}}{s^2+2t^2} \\
&= \frac{s}{s^2+2t^2} - \frac{t}{s^2+2t^2}\sqrt{2}
\end{aligned}$$

Therefore, since $\mathbb{Q}$ is closed under addition, multiplication, and division, we know $\frac{s}{s^2+2t^2} \in \mathbb{Q}$ and $\frac{t}{s^2+2t^2} \in \mathbb{Q}$. Therefore, $\frac{s}{s^2+2t^2} - \frac{t}{s^2+2t^2}\sqrt{2} \in \mathbb{Q}$ and thus $\frac{1}{x} \in K$ if $x \neq 0$. $\square$

Exercise 2.1.15 If $0 < a < b$, prove that:

(a) $a < \sqrt{ab} < b$

Solution:

Proof.

$$\begin{aligned}
ab < b \cdot b &\implies & a \cdot a < ab &\implies \\
ab < b^2 &\implies & a^2 < ab &\implies \\
\sqrt{ab} < \sqrt{b^2} &\implies & \sqrt{a^2} < \sqrt{ab} &\implies \\
\sqrt{ab} < b & & a < \sqrt{ab} &
\end{aligned}$$

Therefore, $0 < a < b \implies a < \sqrt{ab} < b$. $\square$

(b) $\frac{1}{b} < \frac{1}{a}$

Solution:

Proof.

$$
\begin{aligned}
a &< b \\
\frac{a}{1} &< \frac{b}{1} \\
\frac{a}{1} \cdot 1 &< \frac{b}{1} \cdot 1 \\
\frac{a}{1} \cdot \frac{b}{b} &< \frac{b}{1} \cdot \frac{a}{a} \\
\frac{ab}{b} &< \frac{ab}{a} \\
\frac{ab}{b} \cdot \frac{1}{ab} &< \frac{ab}{a} \cdot \frac{1}{ab} \\
\frac{1}{b} &< \frac{1}{a}
\end{aligned}
$$

Therefore, $0 < a < b \implies \frac{1}{b} < \frac{1}{a}$. □

Chapter 2.2: Absolute Value and the Real Line

Exercise 2.2.9 Find all values of x that satisfy the following inequalities. Sketch all graphs.

(a) $|x-2| \leq x+1$

Solution:

$$\begin{aligned} |x-2| &\leq x+1 & &\implies \\ x-2 &\leq x+1 & x-2 &\leq -(x+1) \\ & & x-2 &\leq -x-1 \\ & & 2x &\leq 1 \\ & & x &\leq \frac{1}{2} \end{aligned}$$

Therefore, the inequality holds whenever $x \in (-\inf, \frac{1}{2}]$.

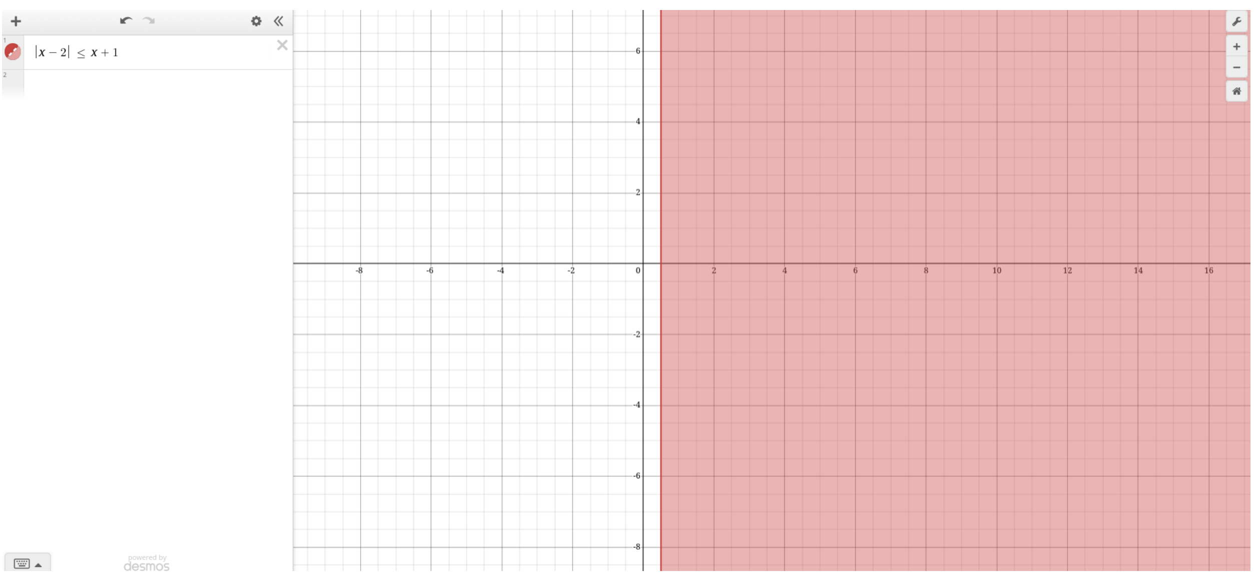

Figure 1: $|x-2| \leq x+1$ as graphed by Desmos.

(b) $3|x| \leq 2-x$

Solution:

$$\begin{array}{lll} 3|x| & \leq 2 - x & \implies \\ 3(x) \leq 2 - x & 3x \leq -(2-x) & \implies \\ 4x \leq 2 & 3x \leq 2 + x & \implies \\ x \leq 12 & 4x \geq 2 & \end{array}$$

Therefore, the inequality holds whenever $x \in (-\infty, \frac{1}{2}] \cup [1, \infty)$.

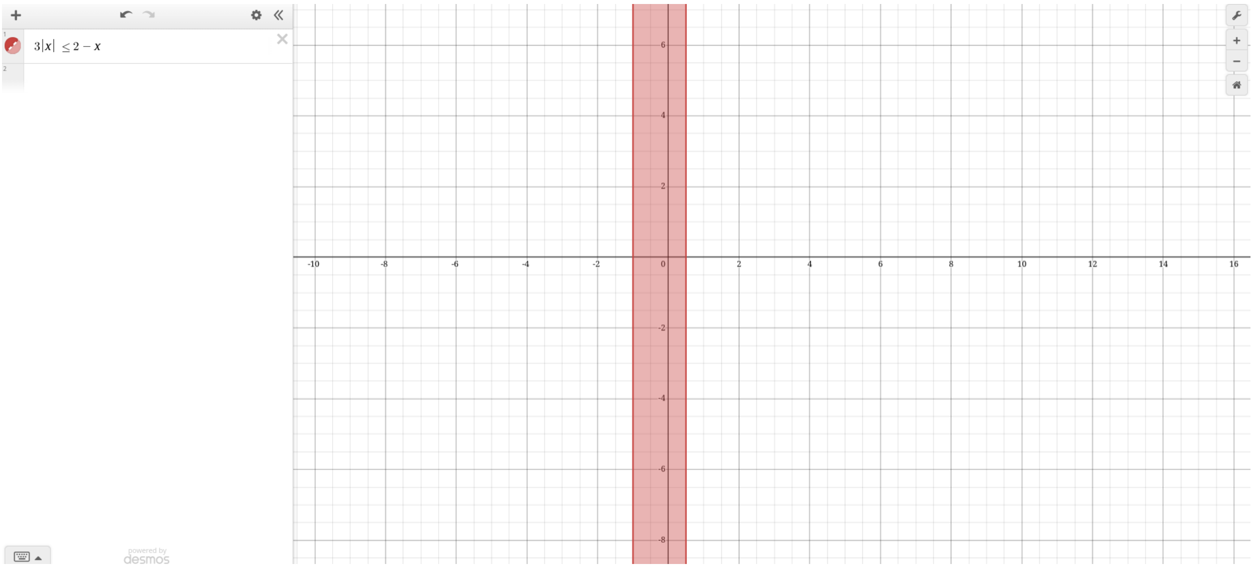

Figure 2: $3|x| \leq 2 - x$ as graphed by Desmos.

Exercise 2.2.14 Determine and sketch the set of pairs $(x, y) \in \mathbb{R} \times \mathbb{R}$ that satisfy:

(a) $|x| = |y|$.

Solution:

$$\begin{gathered} |x| = |y| \implies \\ (x = y \ \vee \ x = -y \ \vee \ -x = y \ \vee \ -x = -y) \implies \\ x = y \ \vee \ x = -y \end{gathered}$$

(b) $|x| + |y| = 1$.

(c) $|xy| = 2$.

(d) $|x| - |y| = 2$.

Exercise 2.2.19 Show that if $a, b, c \in \mathbb{R}$, then the "middle number" is $\text{mid}\{a, b, c\} = \min\{\max\{a, b\}, \max\{b, c\}, \max\{c, a\}\}$.

Solution:

Chapter 2.3: The Completeness Property of $\mathbb{R}$

Exercise 2.3.1 Let $S_3 = \{\frac{1}{n} : n \in \mathbb{N}\}$. Show that sup$S_3 = 1$ and inf$S_3 \geq 0$. (It will follow from the Archimedean Property in Section 2.4 that inf$S_3 = 0$.)

Solution:

(I) Show that sup$S_3 = 1$

By definition of supremum, we know that 1 is the supremum of S_3 if, for all $\epsilon > 0$:

$$1 \geq s_3 \; \forall s_3 \in s_3 \text{AND} \exists s_3 \in S_3 \text{such that} s_3 > 1 - \epsilon.$$

Without appealing to the archimedean property, we know that 1 must be the supremum of this set because

(II) Show that inf$s_3 \geq 0$.

Since $n \in \mathbb{N}$, we know $n \geq 0$ by the order limit properties of $\mathbb{R}$ that $\frac{1}{n} \geq 0$ for all $n \in \mathbb{N}$. Now, let's assume for the purposes of contradiction that $s_1 = \text{inf}S_3 < 0$. In such a case, we know that there exists an $s_2 \in \mathbb{R}$ such that $s_1 > s_2 > 0$. However, we know that s_2 must also lower bound S_3 since $\frac{1}{n} > 0 \; \forall n \in \mathbb{N}$. Thus, any $s_1 > 0$ cannot be the infimum of S_3 since there will always exists a number larger than it (s_2) which lowerbounds S_3. Therefore, by contraduction, the infimum of S_3 must be greater than or equal to zero.

Exercise 2.3.6 Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $\text{inf}(S) = -\text{sup}(\{-s : s \in S\})$.

Solution:

Proof. We can appeal to the completeness property of $\mathbb{R}$ to assert that inf(S) must exist since S is bounded below. Let $m = \text{inf}(S)$. By definition of infiumum, we know that for all $s \in S$, $m \leq s$. Therefore, we know that $-m \geq -s$. Therefore, for the set $K = \{-s : s \in S\}$, $-m$ upperbounds K. Now we must prove that $-m$ is the supremum of K. Since m is the infimum of S, we know that $\forall \epsilon > 0$, $\exists \; s \in S$ such that $m + \epsilon > s$. Therefore, we have $-m - \epsilon < -s$. Therefore, since m upperbounds K, and for arbitrary $\epsilon > 0$, there always exists a $k \in K$ such that $-m - \epsilon < k$, $-m$ must be the supremum of $K = \{-s : s \in S\}$. Therefore, appealing to our earlier definition of m, we have:

$$\begin{aligned} m &= m \\ m &= -(-m) \\ \text{inf}(S) &= -\text{sup}(K) \\ \text{inf}(S) &= -\text{sup}(\{-s : s \in S\}) \end{aligned}$$

□

Exercise 2.3.10 Show that if A and B are bounded subsets of $\mathbb{R}$, then $A \cup B$ is a bounded set. Show that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Solution:

We separate our proof into two parts.

(I) Prove that $A \cup B$ is bounded.

Proof. By definition of boundedness, we know that $\exists\ M_1, M_2 \in \mathbb{R}$ such that $|a| \leq M_1, |b| \leq M_2 \forall a, b \in A, B$. Let $M = max M_1, M_2$. Thus, we trivially have $M \geq M_1$ and $M \geq M_2$. We know that $\forall c \in (A \cup B)$, $c \in A \vee c \in B$. If $c \in A$, then $|c| \leq M$ since $M \geq M_1$ and if $c \in B$, $|c| \leq M$ since $M \geq M_2$. Thus, $A \cup B$ must be bounded for bounded A, B. □

(II) Prove that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Proof. Appealing to the completeness property of $\mathbb{R}$, since A and B are both bounded, we know that $\sup(A)$, $\sup(B)$ both exist. Let $s_1 = \sup(A)$ and $s_2 = \sup(B)$, and $\epsilon > 0$ be arbitrary. By definition of suprema, we know that $\exists a_0, b_0 \in A, B$:

$$s_1 - \epsilon < a_0 \quad \text{and} \quad s_2 - \epsilon < b_0$$

Now, let $s_3 = \sup(\{s_1, s_2\})$. Since the set $\{s_1, s_2\}$ is finite, we know that the supremum must be an element of the set and thus either $s_3 = s_1$ or $s_3 = s_2$ and therefore $s_3 \geq s_1$ and $s_3 \geq s_2$. Let $c \in A \cup B$. Since $c \in A \vee c \in B$, we know trivially that $c \leq s_3$ meaning s_3 upperbounds $A \cup B$. To show that s_3 must be the supremum, consider again $\epsilon > 0$. From the properties of s_1 and s_2 that we discussed earlier, we know that if $s_3 = s_1$, $\exists a_0 \in A$ such that $s_3 - \epsilon < a_0$ and if $s_3 = s_2$, $\exists b_0 \in B$ such that $s_3 - \epsilon < b_0$ for all ϵ. Since $a_0, b_0 \in A \cup B$, there always exists a $c_0 \in A \cup B$ such that:

$$s_3 - \epsilon < c_0$$

Thus, by definition of supremum, we know $s_3 = \sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$. □

Chapter 2.4: Application of the Supremum Property

Exercise 2.4.1 If $S \coloneqq \{\frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}\}$, find $\inf(S)$ and $\sup(S)$.

Solution:

I assert that $\inf(S) = -1$ and $\inf(S) = 1$ and my reasoning is as follows:

$$0 < \frac{1}{n} \leq 1, \quad 0 < \frac{1}{m} \leq 1, \quad \forall m, n \in \mathbb{N} \implies$$

$$\frac{1}{n} - \frac{1}{m} \geq 0 - 1 \qquad\qquad \frac{1}{n} - \frac{1}{m} \leq 1 - 0$$

$$= -1 \qquad\qquad = 1$$

Therefore, -1 and 1 clearly upper and lowerbound S respectively.

To prove that -1 and 1 are the infimum and supremum of S, consider $\epsilon > 0$. Now consider $1 - \epsilon$. By the Archimedean principle, we know that there exists an $n \in \mathbb{N}$ such that $N > 1 - \epsilon$ and

Chapter 2.5: Intervals